Deep Learning for Panoramic Dental Radiographs: A Comprehensive Survey on Caries Detection, Periodontal Bone Loss Assessment and Explainable AI

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***Abstract*—Dental panoramic radiographs (OPGs) are the primary clinical tool for diagnosing caries and periodontal disease, yet manual interpretation is slow and varies substantially between examiners. Deep learning has advanced considerably, but most published systems address a single diagnostic task in isolation, leaving the clinical integration problem unsolved.**

This paper surveys 70 deep learning studies published between 2021 and 2026, covering caries detection, severity staging, periodontal bone loss assessment, tooth segmentation, and explainable AI. Architectures examined include CNNs, YOLO-based detectors, U-Net variants, transformer-based models, and, most recently, vision-language models. We formally define a unified multi-task dental AI framework that combines detection, localization, severity staging, quantification, and explainability within a single system.

Most studies address individual sub-tasks well, and a small number of very recent large-scale vision-language systems trained on proprietary multi-hospital data now integrate detection, tooth numbering, and free-text report generation into one deployed pipeline. However, no publicly reproducible system (open in both model weights and training data) combines detection, quantitative severity staging, and periodontal bone-loss measurement within a single panoramic-radiograph pipeline. Five recurring gaps are identified: task fragmentation across the openly reproducible literature, absent severity staging, low explainability adoption, dataset bias, and limited clinical integration.

The analysis provides a structured foundation for building the next generation of clinically deployable dental AI systems.

***Index Terms*—Deep Learning, Dental Caries, Periodontal Disease, Panoramic Radiographs, Tooth Segmentation and Detection, Multi-Task Learning, Explainable AI**

1. Introduction

Dental panoramic radiography, also known as OrthoPantomography (OPG), is the standard imaging tool in routine dental practice. A single panoramic exposure captures the full dentition, surrounding alveolar bone, and anatomical landmarks in one image, making it the primary modality for detecting dental caries, periodontal bone loss, impacted teeth, and other oral pathologies. Accurate interpretation nevertheless requires substantial expertise and is subject to inter-examiner variability, a problem that affects early-stage practitioners such as dental students and interns more acutely than experienced clinicians.

Dental caries, defined as the progressive bacterial demineralization of tooth enamel and dentine, affects approximately 2.3 billion people globally and is the most prevalent non-communicable disease worldwide [1]. Periodontal disease, characterized by irreversible alveolar bone loss around the teeth, affects over 19% of the adult global population and is the leading cause of adult tooth loss [2]. Both conditions share a clinically consequential property: their early radiographic signs are subtle and easily missed. A faint radiolucency at the proximal surface of a premolar, or a 1–2 mm reduction in alveolar bone height, can alter the clinical decision yet remain visually ambiguous even for trained examiners.

Over the past five years, deep learning has produced notable gains in automated dental radiograph analysis. Convolutional neural networks (CNNs), YOLO-based object detectors, U-Net segmentation architectures, and transformer-based models have demonstrated competitive performance across individual diagnostic tasks such as caries detection, tooth segmentation, and periodontal bone loss estimation [3]–[5].

These advances have not, however, resolved the fragmentation that characterizes the field at the system level. Most published approaches target isolated tasks or single-disease objectives; the components required for a complete clinical workflow are rarely combined. The underlying problem is structural: models are tuned for task-specific metrics rather than the kind of holistic diagnostic reasoning that clinical use demands [6], [7].

Automated dental radiograph analysis is multi-dimensional by nature. A clinically viable system must handle multi-instance detection across all teeth, identify low-contrast early-stage lesions, measure anatomical landmarks such as the cemento-enamel junction (CEJ) and alveolar bone crest, and manage the co-occurrence of multiple diseases within a single panoramic image. This diagnostic process can be conceptualized as a pipeline, as illustrated in Fig. 1.

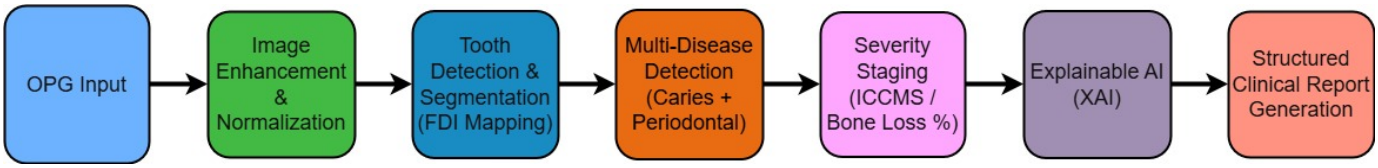


Fig. 1. System Pipeline of Proposed AI Model.

The absence of standardized datasets and evaluation protocols further limits reproducibility and cross-study comparability. A large proportion of studies rely on private, single-center data, which introduces dataset bias and reduces generalization to external clinical environments [7], [8]. Explainable AI (XAI) is also rarely implemented, despite its recognized role in building clinical trust, supporting regulatory acceptance, and enabling interpretable output [9], [10]. Without explainability, combined with inconsistent validation practices, clinical deployment remains out of reach for most published systems.

Rather than surveying individual tasks or model architectures in isolation, this work takes a system-level perspective, examining how individual components might be combined into a unified diagnostic framework [11], [12].

The survey covers 70 papers published between 2021 and 2026, organized into six functional categories: caries detection and severity classification, periodontal disease and bone loss analysis, multi-task end-to-end diagnosis, explainable AI, dataset benchmarking and clinical validation, and tooth segmentation and numbering. The central finding is narrower than it would have been eighteen months ago: a small number of 2026 vision-language systems, most notably DentFound [13], now combine tooth detection, FDI numbering, caries and periodontal findings, post-hoc explainability, and free-text report generation in one deployed system validated on over 101,000 patients. What no reviewed system provides – DentFound included – is a *publicly reproducible* pipeline (open model weights and open training data) that also performs quantitative severity staging and periodontal bone-loss measurement rather than qualitative narrative findings. The unmet need this survey establishes is therefore reproducibility and measurement granularity within an integrated pipeline, not the mere existence of an integrated pipeline.

The main contributions of this survey are as follows:

- A task-level taxonomy of deep learning systems for panoramic dental radiograph analysis, covering detection, segmentation, classification, and multi-task architectures.
- A formal problem formulation for unified multi-disease dental diagnosis, highlighting the integration requirements of detection, severity staging, quantification, and explainability.
- A cross-category comparative analysis of 70 studies, identifying inconsistencies in datasets, evaluation metrics, and validation protocols.
- A quantified gap analysis revealing key limitations in current research, including single-disease model focus, absence of automated severity staging, dataset fragmentation, and low adoption of explainable AI.
- A research roadmap toward clinically deployable, explainable, multi-task dental AI systems.

A unified, explainable, multi-task approach to dental AI is the logical next step toward systems that can be deployed in clinical practice, particularly for supporting junior clinicians and dental students.

2. Search Methodology

This survey used a structured keyword search across five digital libraries: IEEE Xplore, PubMed/MEDLINE, Scopus, Google Scholar, and SpringerLink, restricted to papers published from 2020 onward. The search was conducted in three rounds: an initial search between January and December 2024; a first update extending coverage to March 2025; and a final update, focused on arXiv, Nature Portfolio, MDPI, and NeurIPS/MICCAI proceedings venues, conducted in August 2026 to capture the most recent vision-language and benchmark systems before submission. All results reported in this survey reflect the corpus as of the August 2026 update.

2.1 Search Strategy

Search queries were organized into five thematic clusters:

- *Caries detection on panoramic radiographs:* "dental caries" panoramic radiograph deep learning; "caries detection" panoramic radiograph CNN; "caries segmentation" orthopantomogram neural network.
- *Periodontal disease and bone loss:* "periodontal disease" panoramic radiograph deep learning; "alveolar bone loss" orthopantomogram AI; "CEJ" "bone crest" panoramic radiograph segmentation.
- *Multi-disease and end-to-end OPG systems:* "multi-task" dental panoramic radiograph deep learning; "dental disease detection" panoramic radiograph end-to-end.
- *Explainable AI in dental imaging:* "explainable AI" dental radiograph; "Grad-CAM" dental panoramic; "interpretable deep learning" dental imaging.
- *Dataset and clinical validation:* "dental panoramic radiograph dataset" deep learning; "OPG benchmark" AI; "clinical validation" dental AI panoramic.

2.2 Inclusion and Exclusion Criteria

Papers were included if they: (i) applied deep learning or machine learning to dental radiograph analysis; (ii) used panoramic, bitewing, or periapical radiographs as primary input; (iii) targeted detection, segmentation, classification, or

staging of dental caries, periodontal disease, or related pathologies; and (iv) reported quantitative performance metrics. Papers were excluded if they focused exclusively on CBCT/3D imaging, addressed only non-dental imaging, or did not report reproducible quantitative results. Clinical reviews without AI models were included only when they provided foundational clinical definitions directly relevant to the survey scope.

2.3 Final Corpus

After applying the criteria, 70 papers were selected, spanning 2021–2026: 19 from 2026 (27.1%), 31 from 2025 (44.3%), 17 from 2024 (24.3%), 2 from 2023 (2.9%), and 1 from 2021 (1.4%). This distribution reflects the field's rapid growth over recent years, and in particular a sharp acceleration in 2026 driven largely by vision-language and multimodal-benchmark systems. Of the original 47-paper base corpus (collected through March 2025), publication venues included BMC Oral Health (8 papers), IEEE (8), MDPI journals (9), Elsevier (4), Springer (3), and Scientific Reports/Nature Portfolio (3). The 23 papers added in the August 2026 update skew further toward Nature Portfolio journals (5, including Scientific Reports, *Nature Biomedical Engineering*, npj Digital Medicine, and Scientific Data), MDPI journals (5), and arXiv preprints (5), reflecting how quickly large vision-language and benchmark work in this space is currently moving to preprint servers ahead of peer review.

3. System-Level Taxonomy and Problem Formulation

3.1 Formal Problem Definition

Panoramic dental radiograph analysis is a multi-component diagnostic problem that cannot be reduced to single-task prediction. In clinical practice, meaningful diagnosis requires coordinated execution of several interdependent processes: localization of teeth, identification of pathological conditions, assessment of disease severity, and quantitative measurement of relevant anatomical parameters. Most deep learning systems in the reviewed literature address only one of these components [1], [6], [12], producing outputs that are diagnostically incomplete and difficult to use in a clinical setting.

A unified dental AI system maps an input OPG to structured diagnostic outputs through six core functional components: detection, localization, severity staging, quantification, explainability, and clinical integration. To our knowledge, no reviewed study in the corpus of 70 papers fully addresses all six components simultaneously *and* reproducibly on panoramic radiographs – the closest approach, DentFound [13], covers detection, localization, explainability, and clinical integration but not quantitative severity staging or bone-loss measurement, and releases neither its weights nor its training data. That gap is the primary motivation for this survey.

These six core components are defined as follows:

- **Detection:** Identification of pathological conditions from radiographic features (carious lesions, periodontal bone loss, periapical pathology, impacted teeth). YOLO-family architectures dominate this task [3], [14]–[17].
- **Localization:** Spatial identification of affected teeth through bounding boxes or polygon segmentation. Tooth-level localization using FDI numbering is present in pipeline systems [4], [15]–[17] but absent in classification-only approaches [18]–[20].
- **Severity Staging:** Classification of disease progression along clinically meaningful scales. For caries: enamel, dentine, pulp involvement [16], [21]. For periodontal disease: Stages I–IV per 2018 AAP/EFP classification [22]. This remains a critically underaddressed component: three C1 papers now provide caries severity differentiation [16], [21], [23], led by Wang and Li's D1–D4 staging transformer [23], and two C2 systems now stage periodontal severity on panoramic-adjacent data [24], [25] – neither yet mapped explicitly onto the 2018 AAP/EFP Stage I–IV taxonomy used in this survey.
- **Quantification:** Measurement of the CEJ–alveolar bone crest distance and percentage bone loss. Four studies address this: Jundaeng et al. achieved mAP50 = 0.995 [4]; Wimalasiri et al. reported ICC = 0.80 [26]; Tian et al. achieved ICC = 0.83 [27]; and Le et al. reported per-tooth-type PCC/ICC agreement for radiographic bone loss from a six-keypoint pipeline [24], the most granular quantification result in the corpus.
- **Explainability:** Generation of interpretable visual outputs for clinical verification. 9 of 70 papers now implement some form of explainability (12.9%, up from 6.4% in the pre-2026 corpus): Chen et al. [28] and Bilal et al. [10] use Grad-CAM, Chen et al. [15] and Wang and Li [23] show feature/attention map visualization, and DentFound, George et al., Ren and Chen, Alnasser et al., and Li et al. add Grad-CAM, XGrad-CAM, or CAM-based explanations to 2025–2026 systems [13], [25], [29]–[31].
- **Clinical Integration:** Alignment with diagnostic workflows, reporting standards, and regulatory requirements. Only Schwendicke et al. [32] model economic impact, and only Lee et al. [33] measure clinician acceptance through a 65-dentist usability survey.

This six-component formulation stands in direct contrast to existing approaches that address each task in isolation – a fragmentation independently identified as critical by multiple systematic reviews [2], [6], [11], [12], [34], [35].

3.2 System Taxonomy

Existing approaches can be categorized into four distinct system archetypes based on task integration and diagnostic scope. This taxonomy reveals both progression toward integration and persistent limitations preventing full clinical utility. Table I summarizes capability coverage across taxonomy dimensions.

1) Single-Task Approaches: Single-task systems address one diagnostic objective in isolation – caries detection, tooth segmentation, or periodontal classification – without integration. This accounts for the majority of reviewed studies (21 C1, 18 C2 papers). While several achieve high performance

– CariesXrays FPCL reaching AP50 = 84.1% [15], Attention U-Net achieving Recall = 0.96 [17], U-Net variants reporting Dice = 96.77% [36] – none produce severity grading, periodontal assessment, or clinical explanations. The fundamental limitation is not performance but scope: optimizing one component without modeling interdependence with other diagnostic tasks produces clinically incomplete systems. Systematic reviews confirm this [6], [12], identifying absence of multi-task integration as the primary barrier to clinical translation.

2) Multi-Task Approaches: Fourteen reviewed papers adopt multi-task designs with varying completeness, seven of them added in the August 2026 update (Section 7). Chen et al. [28] is the most clinically complete, classifying both periodontitis and caries on a per-tooth basis using EfficientNet-B0 with YOLOv7 localization (ROC AUC = 98.67% for periodontitis, 98.31% for caries) with Grad-CAM explainability. However, this system operates exclusively on periapical radiographs – not panoramic OPGs – limiting direct applicability. Multi-task approaches expose a critical performance asymmetry: structural tasks (segmentation, enumeration) consistently achieve near-ceiling performance, while pathological tasks (disease detection, severity staging) remain substantially harder. Mureşanu et al. [7] quantified the real-world cost: YOLOv8 F1 dropped from 0.60 (internal) to 0.47 (external) – a 22% degradation from single-center training.

3) Pipeline-Based Approaches: Pipeline systems decompose diagnosis into sequential stages, reflecting the staged clinical reasoning dentists employ. Jundaeng et al. [4] implemented a precise periodontal pipeline: YOLOv8 Stage 1 for tooth segmentation (mAP50 = 0.92) feeds Stage 2 for CEJ and bone crest detection (mAP50 = 0.995, Sensitivity = 1.0). Pornprasertsuk-Damrongsri et al. [17] combined YOLOv5s for tooth detection with Attention U-Net for caries segmentation (Recall = 0.96, F1 = 0.85). The inherent limitation is error propagation: inaccuracies in early stages compound through subsequent stages. None of the reviewed pipelines address severity staging, and explainability appears in only one system [28] – operating on periapical, not panoramic, radiographs.

4) End-to-End Approaches: End-to-end models directly map radiographs to diagnostic outputs using deep architectures – primarily transformers. Bhoopalan et al. [5] applied Vision Transformer and Swin Transformer to 5,098 OPGs, achieving ViT accuracy = 96% without spatial localization or severity grading. Van Nistelrooij et al. [37] combined Mask DINO with Swin-B Transformer, achieving tooth-level assessment performance (FROC = 0.988) but with 40-second inference time unsuitable for clinical use. The fundamental barrier has long been data: the field lacked large-scale, multi-disease, multi-center datasets required for end-to-end learning. DentFound [13] shows this barrier is surmountable given sufficient (private) institutional data – over 101,000 patients across four centers – while MMOral [38] shows a public dataset at meaningful scale (20,563 radiographs) is now available, though not yet paired with an open, end-to-end trained model of DentFound's scope.

The system-level taxonomy and capability gaps across different architectural paradigms are illustrated in Fig. 2

3.3 Task Hierarchy

Dental radiograph analysis follows a structured sequence of interdependent processing stages reflecting the actual cognitive process trained clinicians employ:

Early stages focus on anatomical structure identification, where systems perform well: tooth segmentation achieves pooled sensitivity = 0.92, specificity = 0.94 [35]; tooth enumeration reaches mAP50 = 0.978 [3]. Performance drops in the middle stages – disease detection and severity assessment. Detection on OPG reaches only mAP50 $\approx$ 0.61 [3], and severity staging appears in only three of 21 C1 papers [16], [21], [23]. The terminal stages – explainability and structured reporting – were, until 2026, the most neglected: explainability appeared in only 3 of 47 base-corpus papers (6.4%) [10], [15], [28], rising to 9 of 70 (12.9%) with the August 2026 update, and structured report generation was absent from every system until DentFound and Balel et al. [13], [44] demonstrated it in 2026, albeit on private data and without independent reproducibility for DentFound. Dental AI is therefore still substantially a sequential reasoning problem, with early-stage failures propagating into downstream decisions. For systems targeting junior clinicians, the absence of visual explanations removes any educational value.

3.4 System-Level Limitations

Five system-level limitations persist across the reviewed literature, reflecting field-wide methodological choices:

- **Task Fragmentation:** No panoramic OPG system in the base corpus simultaneously addresses caries detection, severity staging, periodontal quantification, and explainability. Chen et al. [28] achieves dual-disease detection with explainability – closest to unified among the base corpus – but only on periapical radiographs. DentFound [13], identified in the August 2026 update, narrows this gap further on panoramic radiographs by combining detection, periodontal findings, and post-hoc explainability in one system, but still without severity staging, bone-loss quantification, or public model weights and training data.
- **Severity Staging Deficit:** Asadi [21], Bizjak and Robic [16], and, from the 2026 update, Wang and Li [23] provide caries severity differentiation among 21 C1 papers. For periodontal disease, the 2018 AAP/EFP Stages I–IV [22] are not addressed by any system using that exact taxonomy, though Le et al. and Li et al. [24], [25] now perform related periodontal staging outside strict AAP/EFP terms.
- **Explainability Absence:** 9 of 70 papers now implement explainability (12.9%, up from 6.4% in the pre-2026 corpus), still contradicting field consensus [2], [3], [9], [43], [45]. The EU AI Act and FDA guidance require transparency for high-risk systems – requirements 61 of 70 systems still cannot satisfy.

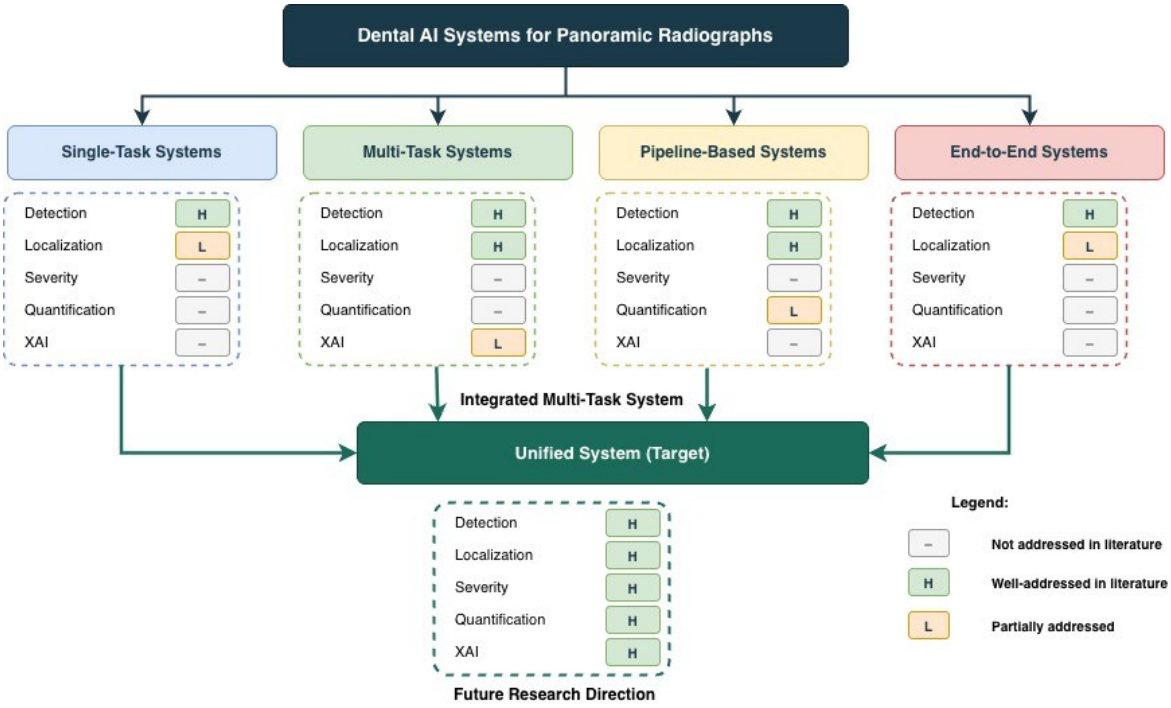


Fig. 2. System-level taxonomy of dental AI approaches for panoramic radiographs, illustrating four architectural paradigms (Single-Task, Multi-Task, Pipeline-Based, and End-to-End Systems) and their respective capability coverage across detection, localization, severity, quantification, and explainability (XAI) dimensions, alongside the proposed Unified System as a future research direction.

TABLE I
SYSTEM TAXONOMY — REPRESENTATIVE STUDIES BY DESIGN PATTERN AND CAPABILITY COVERAGE

System Type	Representative Studies	Detect	Localize	Severity	Quant.	XAI	Multi-Disease
Single-Task	[14], [15], [20], [39]–[42]	✓	Partial	×	×	×	×
Multi-Task	[3], [5], [7], [28], [37], [43]	✓	✓	×	×	Partial	Partial
Pipeline-Based	[4], [15]–[17]	✓	✓	×	Partial	×	×
End-to-End	[5], [37]	✓	Partial	×	×	×	Partial
Vision-Language (2026)	[13]	✓	✓	×	×	Partial	✓
Unified (Required)	*None openly reproducible in reviewed corpus*	✓	✓	✓	✓	✓	✓

✓ = fully addressed; × = not addressed; Partial = partially addressed.
Quant. = quantitative measurement; XAI = explainable AI outputs.
DentFound integrates 4 of 6 dimensions but lacks severity/quantification and open weights/data.

- **Dataset Fragmentation:** Approximately 60% of the original base-corpus studies use private, single-center datasets causing heterogeneity values $I^2 = 88$–96% [12], [34], [35]. The 2026 update's largest systems remain predominantly private as well, though public resources such as MMOral [38] are beginning to emerge. Mureşanu et al. [7] documented 22% F1 degradation from internal to external data. No large-scale, multi-disease, multi-center, publicly available panoramic dataset with standardized annotations exists.
- **Clinical Workflow Disconnection:** Except for Schwendicke et al. [32] (cost-effectiveness analysis) and Lee et al. [33] (65-dentist usability study), no system evaluates AI within actual clinical workflows. Most produce bounding boxes without structured reports or treatment recommendations.

4. BACKGROUND

4.1 Panoramic Radiography in Clinical Dentistry

Orthopantomography (OPG) produces a two-dimensional tomographic representation of the curved dental arches on a flat image plane. Where periapical or bitewing radiographs cover only limited regions, a single panoramic image captures all 32 permanent teeth, maxilla and mandible, and

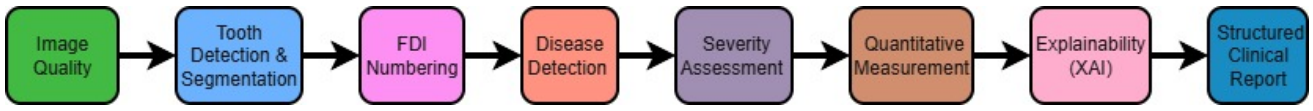


Fig. 3. Task hierarchy for panoramic dental radiograph analysis, illustrating the sequential stages from anatomical structure extraction to clinical report generation. Early stages such as tooth segmentation and numbering achieve high performance, while downstream stages including severity assessment, quantification, and explainability remain underexplored, highlighting key limitations in current systems.

surrounding structures, making it the most information-dense single radiographic acquisition in routine dental practice. For caries detection, panoramic radiographs are less sensitive than bitewing for early proximal lesions [16], [17], [41]. For periodontal bone loss assessment, OPG is the standard screening tool, enabling visualization of the alveolar crest relative to the cemento-enamel junction (CEJ) across all teeth [2], [4], [34]. This modality distinction directly shapes the architectural requirements for AI systems addressing each task.

4.2 Deep Learning Architectures

The architectures reviewed fall into four functional families. First, CNNs with transfer learning – ResNet [40], DenseNet [18], [18], [20], VGG [19], EfficientNet [28], Inception [42], [43] – are used primarily for classification, leveraging ImageNet pre-training to compensate for limited dental dataset sizes. Second, YOLO-family one-stage object detectors (YOLOv5–YOLOv9) [3], [7], [14], [16], [17] dominate detection tasks for their real-time speed and strong bounding box localization. Third, U-Net and nested variants [6], [39], [46]–[48] are the standard for pixel-level segmentation, using skip connections to preserve spatial detail. Fourth, transformer architectures – ViT [5], Swin Transformer [37], [49], RT-DETR [50] – represent the most recent frontier, offering global attention at higher computational cost.

4.3 Evaluation Metrics

A persistent challenge is metric inconsistency, which prevents direct cross-study comparison [2], [6], [34], [35]. Object detection studies report mAP at IoU thresholds of 0.50 and 0.50:0.95. Segmentation studies use IoU and Dice coefficient. Classification studies report accuracy, AUC, sensitivity, specificity, and F1. Some studies [10], [28], [42] also report PPV and NPV, which have direct clinical relevance that aggregate accuracy scores can obscure. The absence of a standardized dental AI evaluation protocol is one of the most frequently cited methodological gaps across the reviewed corpus [1], [2], [34].

5. Caries Detection and Severity Classification (C1)

Caries detection is the most extensively studied task in dental AI, accounting for 21 of the 70 reviewed papers. The studies span object detection, segmentation, classification, and systematic review methodologies across panoramic, bitewing, periapical, and intraoral RGB modalities.

5.1 YOLO-Based Detection on Panoramic Radiographs

Ayhan et al. [14] proposed a two-stage pipeline targeting caries beneath fixed dental prostheses (FDPs), a scenario where metal artifacts from crowns and bridges actively obscure lesion visibility. YOLOv7 detected FDP regions (mAP = 0.968, F1 = 0.956) in Stage 1; a CBAM-enhanced YOLOv7 applied to cropped ROIs then achieved caries detection with mAP = 0.846 and F1 = 0.830 in Stage 2. The CBAM attention module produced measurable gains on low-contrast lesion boundaries beneath metallic artifacts.

Bizjak and Robic [16] applied a two-step YOLOv8 pipeline on the public DENTEX dataset, achieving tooth enumeration mAP@0.5 = 0.95 and lesion sensitivity = 0.70 (internal), 0.67 (external). That external evaluation is among the few genuine cross-site validations in the corpus and confirms the performance drop that single-center training typically produces. Asadi [21] applied Faster R-CNN to the UFBA-UESC public dataset, achieving 94% overall accuracy but with a notable clinical weakness: recall for mild caries was 0.64 versus 0.69 for severe caries – the inverse of clinical need, where early lesions matter most. Pornprasertsuk-Damrongsri et al. [17] combined YOLOv5s for tooth detection with Attention U-Net for caries segmentation on 500 panoramic radiographs with bitewing-confirmed ground truth, achieving Recall = 0.96, F1 = 0.85 with Bland-Altman agreement analysis. An example of multi-tooth carious lesion detection with severity differentiation in panoramic radiographs is shown in Fig. 4.

5.2 U-Net Segmentation Approaches

Alharbi et al. [39] compared U-Net, U-Net++, and U-Net3+ on 1,500 panoramic images. U-Net3+ achieved the best test performance (IoU = 0.67, Dice = 0.60, Accuracy = 0.95); its full-scale skip connections gave a measurable advantage over standard U-Net (Dice = 0.58). The gap between training IoU (0.77) and test IoU (0.67) reflects overfitting driven by class imbalance – only 510 of 1,500 images contained visible cavities.

Lim et al. [47] proposed Faster R-CNN + U-Net (EfficientNet-B0 encoder) on 597 panoramic radiographs, achieving IoU = 0.608, Dice = 0.743, Recall = 0.731 on 33,752 augmented cropped tooth images, with no severity grading or explainability. Asci et al. [46] applied U-Net to 6,075 pediatric panoramic radiographs across primary, mixed, and permanent dentition, achieving F1 = 0.8675, Sensitivity = 0.8269, Precision = 0.9123. The results show U-Net's adaptability across diverse tooth morphologies, though AUC = 0.76 falls below the 90% clinical threshold. Vickram and Mathan Muthu [36] applied UNet variants to 500 dental X-ray images, achieving

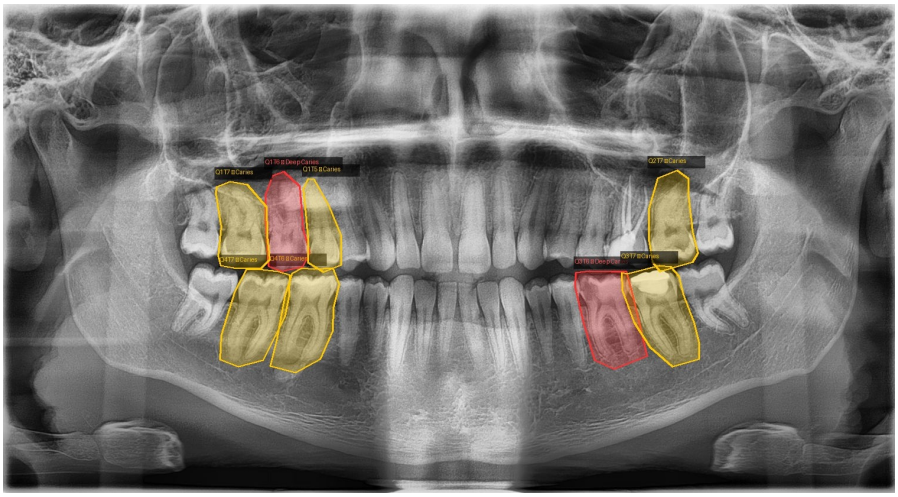


Fig. 4. Example panoramic radiograph with annotated carious lesions, illustrating multi-instance detection and severity differentiation (caries vs. deep caries) across multiple teeth.

Accuracy = 98.03% and Dice = 96.77%, though on a small dataset with no external validation.

5.3 CNN Classification and Transfer Learning

Zancan et al. [42] applied InceptionResNet-v2 to 7,304 tooth-level images from 935 panoramic radiographs, achieving accuracy = 0.94 and F1 = 0.94 with five-fold cross-validation – among the more rigorous validation schemes in C1. Kaushik and Sharma [18] applied DenseNet201 to 29,815 dental X-ray images, reporting 91% overall accuracy alongside a cavity-specific F1 of only 0.27, which illustrates how class imbalance can render aggregate accuracy misleading. Ozdemir et al. [20] applied DenseNet-121 to 14,498 cropped tooth images (2,303 caries / 12,195 non-caries), achieving accuracy = 93.17% and F1 = 87.49%, though without localization on full panoramic images. Khuwaja et al. [40] evaluated YOLOv8, ResNet-50, EfficientDet, and MobileNetV2 on 7,447 intraoral RGB smartphone images targeting mobile diagnostics in resource-limited settings, with ResNet-50 achieving mAP@0.50 = 0.670.

5.4 Systematic Reviews and Meta-Analyses

Albano et al. [12] reviewed 20 studies via PRISMA/QUADAS-2, finding accuracy 73%–98% and AUC 0.84–0.98 across CNN-based models. High heterogeneity was a consistent finding, and the authors called for standardized datasets and multicenter validation. Luke and Rezallah [1] conducted a meta-analysis of 21 studies (17,190 total samples); the best individual study achieved 98.8% accuracy, but I^2 values of 88–96% signal heterogeneity severe enough to undermine pooled estimates. Zanini et al. [6] reviewed 42 studies and confirmed that single-task formulations and absent caries severity differentiation are field-wide patterns, not isolated oversights. Sastika and Marappan [45] reviewed accuracy ranges of 85–99% across multiple architectures and explicitly identified Grad-CAM as a requirement in the studies surveyed.

Chen et al. [15] introduced the CariesXrays dataset (6,000 hospital-grade panoramic X-rays, 13,783 expert-verified bounding boxes) alongside the FPCL architecture, achieving AP = 48.2%, AP50 = 84.1%, outperforming YOLOv8 (AP50 = 80.7%). CariesXrays is the most substantial public benchmark contribution in the C1 category.

5.5 Bitewing Caries and Cost-Effectiveness

Perez de Frutos et al. [41] evaluated RetinaNet, YOLOv5-M, and EfficientDet on 13,887 bitewing X-rays from the HUNT4 Oral Health Study, annotated by six dentists with a 197-image consensus test set. YOLOv5-M achieved mAP = 0.647, F1 = 0.548, matching or surpassing individual expert annotators – evidence that deep learning can reach human expert performance on proximal caries detection. Schwendicke et al. [32] combined U-Net performance evaluation (accuracy = 0.80, sensitivity = 0.75 vs. dentist sensitivity = 0.36) with a Markov lifetime cost-effectiveness simulation on 3,686 bitewing radiographs. The analysis projected increased tooth retention (64 vs. 62 years) and reduced cost per tooth (€298 vs. €322), yielding ICER = −13.9 €/year, which places AI caries detection in the cost-dominant category.

5.6 2026 Update: Transformer, Segmentation-Benchmark, and Staging Approaches

An August 2026 literature update identified three further C1 studies not available at the time of the original search. Wang and Li [23] trained a 12-layer hierarchical transformer with a multi-scale feature pyramid on a private three-hospital corpus of 3,856 panoramic radiographs (12,847 lesions), achieving mAP = 87.3% across caries stages with an explicit D1–D4 severity grade, per-grade sensitivity (81.3% for the hardest D1 lesions), specificity = 91.7%, and expert-agreement $\kappa = 0.84$ on a genuine patient-level 70/15/15 held-out test split – one of the few C1 studies in this survey with a true patient-level held-out test set, and a methodological bar most other reviewed caries-severity work, including single-split validation

TABLE II
SUMMARY OF REVIEWED STUDIES — CARIES DETECTION AND SEVERITY CLASSIFICATION (C1)

Ref	Model	Dataset Size	Best Metric	XAI	Key Gap
[14]	YOLOv7 + CBAM	1,004 OPG	mAP=0.846, F1=0.830	None	Single-center, no XAI
[40]	YOLOv8, ResNet-50, EfficientDet, MobileNetV2	7,447 RGB	mAP@0.50=0.670	None	RGB only, no radiographs
[39]	U-Net, U-Net++, U-Net3+	1,500 OPG	IoU=0.67, Dice=0.60	None	Class imbalance, no severity
[46]	U-Net	6,075 OPG (pediatric)	F1=0.8675, AUC=0.76	None	Below 90% clinical threshold
[12]	Systematic Review (CNNs)	20 studies	Acc up to 0.98, AUC 0.98	None	High heterogeneity
[18]	DenseNet201	29,815 X-rays	Acc=91%, Cavity F1=0.27	None	Severe class imbalance
[15]	FPCL (Faster R-CNN + D2D-FPN + Contrastive)	6,000 OPG	AP=48.2%, AP50=84.1%	Attn maps	No severity/periodontal
[41]	YOLOv5-M, RetinaNet, EfficientDet	13,887 bitewing	mAP=0.647, F1=0.548	None	Bitewing only, no periodontal
[45]	Review: U-Net, ResNet, YOLOv7/v8, ensemble	Multiple studies	Acc 85–99%, F1 87–94%	Discussed	No original experiment
[32]	U-Net	3,686 bitewing	Acc=0.80, ICER=−€13.9/yr	None	Single-center, German context
[1]	Meta-analysis (CNNs)	21 studies, 17,190 images	Best Acc=98.8%	None	I^2=88–96% heterogeneity
[36]	UNet, UNet+, UNet++	500 images	Acc=98.03%, Dice=96.77%	None	Small dataset, no clinical val.
[23]	Hierarchical Transformer + FPN	3,856 OPG, 12,847 lesions	mAP=87.3%, D1–D4 staged	Partial	Private, metallic-artifact errors
[51]	12-architecture benchmark (CNN/ViT/Mamba)	DC1000 (597 OPG)	Best Dice=0.7345 (CNN)	None	Small data regime only
[52]	Faster R-CNN + U-Net	3,000 OPG (AI-Hub)	Dice=0.9482 (unverified)	None	Protocol not confirmable (paywalled)
[42]	InceptionResNet-v2	935 OPG / 7,304 teeth	Acc=0.94, F1=0.94	None	No localization, no periodontal
[47]	Faster R-CNN + U-Net (EfficientNet-B0)	597 OPG	IoU=0.608, Dice=0.743	None	No severity, no periodontal
[6]	Systematic Review (U-Net, Mask R-CNN, ResNet)	42 studies	Various (review)	None	No severity differentiation
[16]	YOLOv8 two-step pipeline	1,005 OPG + 85 external	mAP@0.5=0.95, Sens=0.70	None	Caries only, no severity
[21]	Faster R-CNN	1,000 OPG	Acc=94%, Mild recall=0.64	None	Weak mild caries detection
[17]	YOLOv5s + Attention U-Net	500 OPG, 14,997 teeth	Recall=0.96, F1=0.85	None	Single-disease, no periodontal
[20]	DenseNet-121	14,498 tooth images	Acc=93.17%, F1=87.49%	None	No localization, no severity

schemes elsewhere in this corpus, does not meet. Ghimire et al. [51] benchmarked twelve CNN, vision-transformer, and Mamba (state-space) architectures for caries segmentation on the DC1000 dataset under a controlled 497/100 train/test protocol, finding a plain CNN (DoubleU-Net, Dice = 0.7345) outperformed every transformer and Mamba variant tested – a useful ceiling estimate for contextualizing unusually high caries-segmentation Dice scores elsewhere in the literature. Kim et al. [52] reported a two-stage Faster R-CNN + U-Net caries segmentation pipeline on 3,000 panoramic radiographs from the Korean AI-Hub corpus with Dice = 0.9482; this figure is markedly higher than every other caries-segmentation result in this survey, and because the evaluation protocol is behind a publisher paywall it could not be independently confirmed, so it is reported here with that explicit caveat rather than as a verified state of the art.

6. PERIODONTAL DISEASE AND ALVEOLAR BONE LOSS DETECTION (C2)

Periodontal disease detection from radiographs requires quantifying the spatial relationship between anatomical landmarks – specifically the distance between the cemento-enamel junction (CEJ) and the alveolar bone crest – across all teeth in the panoramic image. This task demands precise spatial localization and continuous measurement rather than binary classification. Eighteen papers in the C2 category reflect a less mature but rapidly advancing subfield.

6.1 Clinical Foundation

Chmielewski et al. [22] provide essential clinical context: the 2018 AAP/EFP classification defines periodontitis

stages I–IV and grades A–C based on bone loss percentage, probing depth, tooth loss, and risk factors. Across 11 clinical studies covering 459 cases and 852 examiners, staging accuracy peaked at 90% but inter-examiner agreement remained variable. This is precisely the subjectivity problem that AI systems could address – provided they produce quantitative, staged outputs aligned with the 2018 criteria, rather than a binary bone loss label. Fig. 5 illustrates the spatial nature of periodontal bone loss assessment, which requires accurate identification of landmarks such as the CEJ and alveolar bone crest.

6.2 Landmark Detection and Bone Loss Quantification

Jundaeng et al. [4] reported the most technically complete periodontal AI system in the reviewed corpus. Their dual YOLOv8 pipeline on 2,000 panoramic radiographs achieved tooth segmentation (Accuracy = 0.97, mAP50 = 0.92) and CEJ/bone level detection (Accuracy = 0.98, Sensitivity = 1.0, Specificity = 0.98, F1 = 0.90, mAP50 = 0.995). A sensitivity of 1.0 for bone level detection carries practical weight: missed bone loss in a screening context is more harmful than a false positive. Limitations include single-center data, images of uniformly high quality, and no caries integration or explainability.

Iacob et al. [34] synthesized evidence from 20 diagnostic accuracy studies covering 141,166 radiographs through PRISMA/QUADAS-2. U-Net, YOLO, ResNet, ConvNeXt, and DeNTNet have each been applied, but heterogeneity across studies prevents reliable pooling. Khubrani et al. [2] meta-analyzed 30 studies using APPRAISE-AI, reporting pooled sensitivity = 87%, specificity = 76%, accuracy = 84% – figures that are useful as baselines but insufficient for autonomous clinical use.

Mathur et al. [19] applied VGG-16 to 1,874 panoramic radiographs for binary bone loss classification (accuracy = 89.27%, sensitivity = 83.27%). This approach, the most common in C2, is also the most clinically limited: it produces no staging information. Cerda Mardini et al. [53] attempted quantitative staging on 500 panoramic radiographs, reporting F1 = 0.29 for light-to-moderate bone loss and F1 = 0 for severe bone loss. The results honestly document how difficult quantitative staging is, and the model's failure on the most clinically critical severity class is worth noting. Kong et al. [48] integrated SRGAN super-resolution preprocessing with U-Net for bone loss segmentation (Dice = 0.91, IoU = 84.9%), the strongest segmentation metrics in C2. The results suggest that preprocessing quality directly affects segmentation accuracy, though the 698-image dataset is small and no disease staging is included.

Wimalasiri et al. [26] applied YOLOv8x and Keypoint R-CNN to 1,000 intraoral periapical X-rays, achieving ICC = 0.80 for severity and 87% for bone loss pattern classification (horizontal vs. vertical). Pattern classification is absent from all other reviewed systems and has direct prognostic value for surgical planning. Tian et al. [27] developed the PANPOL mathematical model validated on 290 panoramic patients, achieving ICC = 0.83, though this is a rule-based rather than a deep learning approach.

6.3 Periodontitis Classification

Bilal et al. [10] proposed PDCNET – a custom five-block CNN with Grad-CAM explainability – and evaluated it against EfficientNet-B0, DenseNet-201, VGG-16, VGG-19, Inception-V3, and MobileNet on a Kaggle dataset augmented with SMOTE-Tomek, achieving accuracy = 98.39%, AUC = 99.79%, F1 = 98.31%. These are the highest classification metrics in C2, though the original sample of 50 positive and 50 negative cases raises serious concerns about statistical reliability. The paper is useful for demonstrating the feasibility of Grad-CAM in periodontal AI. Ferrara et al. [9] reviewed 13 periodontal deep learning studies via PRISMA/QUADAS-2, reporting accuracy ranges of 73%–98.6% and noting that explainable AI is absent from every system reviewed. Lee et al. [33] assessed AI alveolar bone loss measurement usability through a survey of 65 dentists on 550 bitewing images – one of the few clinical usability studies in the entire corpus.

Kim et al. [54] proposed an unsupervised few-shot learning architecture combining UNet, CVAE, and clustering methods (k-means/DBSCAN/GMM) on 1,372 panoramic radiographs. The approach achieved a 14% improvement over supervised ViT, suggesting that label-efficient learning may be a practical path forward in data-scarce dental AI settings. Karamuftuoğlu et al. [50] benchmarked YOLOv12x, RT-DETR-L, and RT-DETR-X on a 387-image pediatric panoramic dataset for furcation lesion detection, achieving mAP@0.5 = 0.434. The modest result reflects both the difficulty of the task and the constraints of small single-center datasets.

6.4 2026 Update: Keypoint Staging and Multicenter Validation

Four further C2 studies were identified in the August 2026 update, and they represent the strongest periodontal evidence in the entire corpus. Le et al. [24] proposed a YOLOv11 six-keypoint pipeline for automated periodontitis diagnosis and staging on 607 panoramic radiographs, reporting segmentation precision = 0.981/recall = 0.984/F1 = 0.983/mAP50 = 0.994, keypoint mAP50 $>$ 0.958, and per-tooth-type PCC/ICC agreement for radiographic bone loss – directly answering the quantification gap this survey identifies in Section 3. Li et al. [25] introduced HC-Net+, evaluated on a multicenter cohort with AUROC = 94.2% against a specialist AUROC of 85.6% and AUROC = 0.967 on a second external validation cohort, with CAM-based explainability and specialist agreement $\kappa = 0.64$; the authors also report that 20.6% of Stage II cases were missed even by this system, a clinically important caveat that should temper the headline AUROC. Güner et al. [55] synthesized 80 studies on fractal analysis and AI for radiographic bone loss detection under a QUADAS-2/PROSPERO-registered protocol (PROSPERO 1278324), providing the most current systematic evidence base for this subfield. Lee et al. [56] meta-analyzed 23 further studies comparing binary versus staged periodontitis

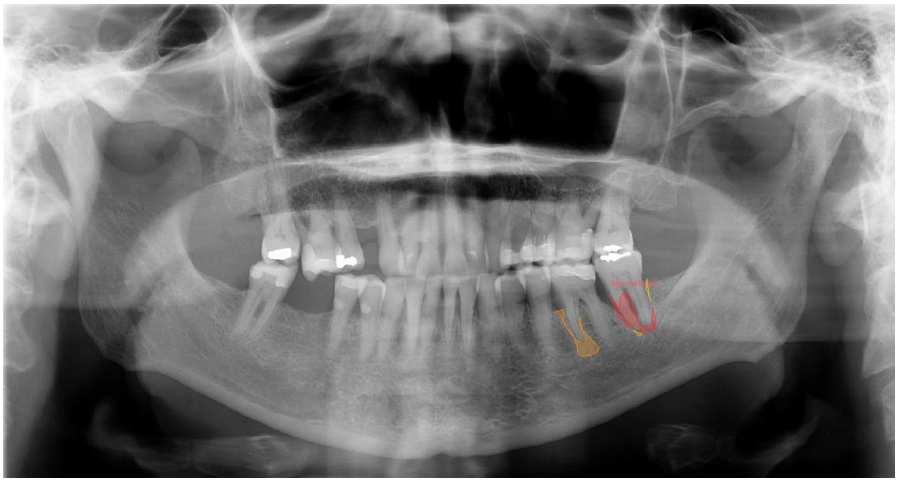


Fig. 5. Example panoramic radiograph illustrating periodontal bone loss regions. Accurate assessment requires spatial measurement between the cemento-enamel junction and alveolar bone crest.

classification across imaging modalities, reporting pooled sensitivity, specificity, and AUC separately by modality and by binary-versus-staging task – direct supporting evidence for this survey's repeated observation that binary detection and true clinical staging are different problems with different, generally lower, performance ceilings for the latter. Together, these three works are the first in this survey's corpus to combine periodontal severity staging with any form of external or multicenter validation, directly narrowing – though not closing – Gap 2 and Gap 4 as defined in Section 14.

7. Multi-Task and End-to-End OPG Diagnosis (C3)

Fourteen papers address multiple diagnostic tasks within a single pipeline applied to panoramic radiographs – the architectural direction most consistent with real clinical utility.

Chen et al. [28] report the most clinically complete dual-disease system in the corpus, combining YOLOv7 for tooth localization with EfficientNet-B0 for simultaneous multi-label classification of periodontitis and dental caries on periapical X-rays, with Grad-CAM visualization. EfficientNet-B0 achieved ROC AUC = 98.67% for periodontitis and 98.31% for dental caries with 10-fold cross-validation. The critical limitation is modality: periapical, not panoramic.

Nassiri and Akhloufi [3] compared YOLO v5–v9 on DENTEX across quadrant segmentation, tooth enumeration, and disease detection. YOLOv9e-seg achieved mAP50 = 0.978 for tooth enumeration and 0.959 for segmentation, while pathology detection peaked at mAP50 = 0.61 – confirming that structural tasks are tractable while pathology detection remains the harder problem.

Van Nistelrooij et al. [37] combined OdontoAI and DENTEX datasets with Mask DINO (ViT backbone) and Swin-B Transformer, achieving FROC = 0.988 for tooth segmentation and AUC = 0.996 for impacted teeth, the strongest tooth-level assessment results reviewed. A 40-second inference time per image is the primary barrier to clinical use. Gamal et al. [43] demonstrated that a weighted average ensemble of CNN architectures (Custom CNN, ResNet-50, InceptionV3, DenseNet121, VGG19) achieves accuracy = 92.93% on 1,814 cropped tooth images. ESRGAN super-resolution preprocessing contributed roughly 3% of that accuracy gain.

Mureşanu et al. [7] provided the most methodologically transparent external validation in C3: YOLOv8 tested on a Romanian clinical dataset showed internal F1 = 0.60 degrading to external F1 = 0.47 – a 22% drop that directly measures the generalization cost of single-center training. Bhoopalan et al. [5] applied ViT and Swin Transformer to 5,098 OPG images, achieving ViT accuracy = 96%, precision = 95.8%, recall = 96.2%. These are strong results for image-level classification, though the system provides neither localization nor segmentation. Zhou et al. [11] reviewed datasets, methodologies, and emerging challenges in dental image analysis, reinforcing the need for standardization across the field.

7.1 2026 Update: Vision-Language Systems and Benchmarks

The August 2026 update identified seven further C3 systems, and they change the framing of this survey's central claim. Zhu et al. [13] introduced DentFound, an instance-guided vision-language model that injects tooth-instance masks as visual embeddings into a transformer-based VLM to produce tooth-level diagnosis and free-text radiology reports across 98 disease categories and 11 post-treatment categories. DentFound was trained and validated on a private multi-hospital corpus exceeding 101,000 patients across four clinical centers, with an expert reader study on 595 panoramic radiographs in which 12 dentists and radiologists rated DentFound-generated reports as superior to or comparable with human-written reports, and with broader diagnostic coverage. It provides post-hoc Grad-CAM-style regional-response explainability tied to individual generated findings and identifies tooth position across 52 sites using FDI-style numbering. This is, to our knowledge, the most clinically integrated dental panoramic system published to date. It is also, on close reading, not the system this survey's original central claim needs to rule

TABLE III
SUMMARY OF REVIEWED STUDIES — PERIODONTAL DISEASE AND BONE LOSS (C2)

Ref	Model	Dataset Size	Best Metric	XAI	Key Gap
[4]	Dual YOLOv8 (teeth + CEJ/bone)	2,000 OPG	mAP50=0.995, Sens=1.0	None	Single-center, no caries
[22]	Clinical review (no ML)	459 cases, 852 examiners	Stage acc=90%	None	No AI system proposed
[34]	Systematic Review (CNN, YOLO, U-Net)	141,166 radiographs	Various (review)	None	High heterogeneity
[19]	VGG-16 (transfer learning)	1,874 OPG	Acc=89.27%	None	Binary only, no staging
[2]	Meta-analysis (CNN, YOLO, U-Net, SVM)	30 studies	Sens=87%, Spec=76%	None	Pooled heterogeneity
[27]	PAN-POL mathematical model	290 PAN patients	ICC=0.83	None	Not deep learning, manual
[33]	Faster R-CNN + DeepLab v3+	550 bitewing	Acc=82–87% (severe)	None	Bitewing only, no OPG
[53]	Hybrid DCNN (Xception) + rule-based PBL	500 OPG	F1=0.29 (lt-mod), F1=0 (severe)	None	Fails on severe PBL
[26]	YOLOv8x + Keypoint R-CNN	1,000 IOPA	ICC=0.80, Pattern Acc=87%	None	Periapical only, no OPG
[48]	SRGAN + U-Net	698 OPG	Dice=0.91, IoU=84.9%	None	No severity, small dataset
[9]	Systematic Review (CNNs, hybrid)	13 studies	Acc 73–98.6%	None	No XAI in any reviewed study
[10]	PDCNET (custom CNN) + Grad-CAM	50+50 orig (SMOTE aug.)	Acc=98.39%, AUC=99.79%	Grad-CAM	Dataset too small
[50]	YOLOv12x, RT-DETR-L, RT-DETR-X	387 OPG (pediatric)	mAP@0.5=0.434	None	Small dataset, modest mAP
[54]	UNet + CVAE + k-means/DBSCAN/GMM	1,372 OPG	+14% over supervised ViT	None	No bounding boxes

out: the accessible portions of the paper report no quantitative caries severity staging (no D1–D4 or equivalent grade) and no numeric periodontal bone-loss quantification (no CEJ-to-crest distance, percentage bone loss, or AAP/EFP stage); its outputs are narrative findings rather than the graded, measurement-level outputs this survey's taxonomy requires in Section 3. Reproducibility is also asymmetric: the inference code is released under an Apache 2.0 license, but neither fine-tuned model weights nor the training data are public, so the system cannot currently be independently re-run or externally audited by a third party. An Author Correction was published on 8 July 2026; readers relying on specific DentFound figures should consult it.

Three concurrent benchmark efforts test whether general-purpose vision-language models can substitute for task-specific pipelines like DentFound's or the one proposed in this survey, and all three answer no. Hao et al. [38] released MMOral, an instruction and benchmark dataset of 20,563 annotated panoramic radiographs (1.3 million instruction-following instances), and evaluated 64 large vision-language models zero-shot: the best proprietary model, GPT-4o, reached only 43.3% accuracy, and a medical-specific model (HealthGPT-XL32) reached only 39.7%; supervised fine-tuning on MMOral lifted a mid-sized open model from 21.5% to 46.2%. Li et al. [57] introduced PanDent, 9,524 expert-validated orthopantomograms for tooth-level structure-language consistency, reporting that current MLLMs produce fluent but clinically inconsistent reports with substantial fine-grained localization errors. Jha et al. [58] introduced DentiAsk, a 10,000-question VQA benchmark spanning descriptive, spatial, and numerical-quantification reasoning tiers, and found that models degrade sharply on exactly the spatial-localization and counting questions that panoramic diagnosis depends on. Taken together, these benchmarks are direct evidence that deterministic, task-specific detection-plus-classification pipelines – of the kind formalized in this survey's taxonomy (Section 3) – remain necessary: general VLMs are not yet a substitute for explicit tooth-level geometric grounding.

Two further 2026 systems are methodologically relevant without matching DentFound's scale. Balel et al. [44] paired YOLOv12 detection/segmentation with locally hosted LLMs (DeepSeek R1, Mistral, Llama 3.2) to generate structured reports from 30,954 panoramic radiographs, and found that commercial cloud LLMs (ChatGPT, Gemini, cloud DeepSeek R1) hallucinated findings in 100% of generated reports, versus 30 hallucinations across 50 reports for the best local, structured-JSON-grounded pipeline – a direct caution for any future report-generation component of a system like ours. Eftimie et al. [59] (published under the name Sorana Muresanu on an earlier preprint) reported a two-stage YOLOv11-then-RT-DETR pipeline that detects, segments, and assigns 14 pathology categories to individual teeth, evaluated on both an institutional

corpus and, as a cross-dataset generalization test, the public Tufts Dental Database – the closest published architectural analogue to the detect-then-classify system design formalized in this survey's taxonomy (Section 3), and one of the only C3 systems to report a working lesion-to-tooth assignment step end to end. Alnasser et al. [31] compared five CNN backbones on a 35-class, 10,580-image panoramic dataset with Grad-CAM explainability, reaching 97.5% whole-image accuracy; because the evaluation uses a random rather than patient-level split on an aggregated public-mirror dataset, near-duplicate leakage cannot be excluded and the figure should be read as an upper bound rather than a clean generalization estimate.

8. Explainable AI in Dental Diagnostics (C4)

Of the original 47-paper base corpus, only three reported any explainability output: Chen et al. [28] use Grad-CAM for dual-disease classification on periapical X-rays; Bilal et al. [10] apply Grad-CAM to periodontal disease classification; and Chen et al. [15] include attention visualization in feature maps for caries detection. That constituted **6.4%** of the base corpus – a figure that directly quantified the explainability deficit in dental AI as of March 2025.

The August 2026 update raises this to **9 of 70 papers (12.9%)**. Wang and Li [23] provide attention visualization maps alongside their caries-staging transformer, though the authors themselves describe formal explainability as future work rather than a validated method. DentFound [13] generates post-hoc regional-response visualizations tied to individual report findings, evaluated qualitatively by clinicians rather than against ground-truth localization. Alnasser et al. [31] apply Grad-CAM across a 35-class classifier, and Li et al. [25] report CAM-based explanations alongside HC-Net+'s periodontitis staging. Two further 2026 papers are explicitly about explainability rather than merely using it: George et al. [29] propose a two-stage YOLOv8-plus-ResNet-50 framework for pericoronitis assessment with Grad-CAM shown to reach 84% concordance with radiologist annotations on a 50-image reader check; Ren and Chen [30] compare five CAM-family methods for caries classification explainability and find XGrad-CAM the strongest, though the modality used is not confirmed to be panoramic and should be treated cautiously in that respect. Mehta et al. [60] synthesize explainability, bias, and generalizability across 11 dental-imaging studies, reporting accuracy figures up to 95.1% alongside accuracy-trustworthiness correlations ranging from 0.367 to 0.987 depending on study design – a useful reminder that high accuracy and high trustworthiness are not the same property and do not covary reliably across this literature. Even at 12.9%, explainability implementation remains a minority practice, and no method beyond the CAM family (SHAP, LIME, or transformer attention-based interpretability) has been applied to panoramic dental diagnosis anywhere in this corpus.

Gradient-weighted Class Activation Mapping (Grad-CAM) generates a coarse localization map of the image regions most influential for a CNN prediction. In dental diagnostics, it serves three practical functions: (i) it allows clinicians to verify that the model attends to anatomically plausible regions; (ii) it can function as a visual teaching tool for junior clinicians learning radiographic interpretation; and (iii) it creates an audit trail relevant to the EU AI Act and FDA guidance on AI/ML-based medical devices, both of which classify AI diagnostic tools as high-risk systems requiring transparency and human oversight [28].

Field consensus holds that explainability is necessary for clinical adoption [3], [9], [43], [45], yet the implementation rate remains under 13%. For systems targeting junior clinicians and dental students, this is more than an oversight – a visual justification is what transforms AI-assisted diagnosis from prediction into education. SHAP, LIME, and transformer attention visualization remain almost entirely unused in dental AI outside a handful of unvalidated attention-map visualizations; transformer attention maps are already intrinsic to the architecture and add no computational overhead [5].

9. Comparative Analysis of Representative Methods

Table IV restricts comparison to the 25 studies that report a single clear headline metric directly comparable across tasks; the newer 2026 systems discussed in Sections 7 and 8 are deliberately excluded here because several of them (DentFound in particular) report reader-study ratings and narrative-generation quality rather than a single quantitative metric, and forcing such results into a single "Best Metric" cell would misrepresent them. Within these 25 studies, task scope splits into 14 single-task, 6 multi-task, 2 multi-stage, and 3 review/survey works – confirming that single-disease, single-task formulations remain the dominant experimental design even among the field's most cited results. Validation practice is similarly concentrated: 19 of the 25 report only internal (same-center) validation, 3 report a genuine external or cross-site cohort ([7], [16], [41]), and 3 are themselves reviews. Explainability is essentially absent from this representative subset, with only 2 studies applying a validated Grad-CAM output and 1 further study showing unvalidated feature-map attention – the remaining 22 provide no interpretability output at all.

Modality is heavily skewed toward panoramic imaging (17 of 25 OPG, with 2 bitewing, 2 periapical/X-ray, 1 intraoral periapical, and 3 multi-modality reviews), which supports treating panoramic radiographs as the primary modality for this survey while noting that comparisons against bitewing- or periapical-only studies elsewhere in this table are not strictly like-for-like. Reported metrics are also inconsistent enough to limit direct comparison: 8 of the 25 studies report an mAP-family detection metric, 7 report accuracy or AUC, 3 report Dice, and the remaining 7 each report a distinct metric (F1, recall, ICC, or sensitivity) used nowhere else in the table. This metric fragmentation, more than any single result, is why Table IV is presented as a qualitative landscape rather than a ranked comparison.

Finally, the table makes the severity-and-quantification gap concrete: of these 25 representative studies, only Asadi [21]

TABLE IV
COMPARISON OF REPRESENTATIVE DEEP LEARNING METHODS FOR DENTAL RADIOGRAPH ANALYSIS. PERFORMANCE METRICS ARE REPORTED AS PROVIDED IN THE RESPECTIVE STUDIES AND ARE NOT DIRECTLY COMPARABLE DUE TO VARIATIONS IN TASKS, DATASETS, AND EVALUATION PROTOCOLS.

Ref	Task	Scope	Dataset (Size)	Modality	Model	Best Metric	XAI	Clinical Val.	Key Limitation
[14]	Caries Detection	Single-task	Private OPG (1,004)	OPG	YOLOv7 + CBAM	mAP=0.846	No	Internal	No external validation
[4]	Bone Loss + Periodontal	Multi-task	Private OPG (2,000)	OPG	YOLOv8	mAP50=0.995	No	Internal	Single-center dataset
[28]	Multi-disease Detection	Multi-task	Periapical (11k)	X-ray	YOLOv7 + EfficientNet	AUC≈98%	Yes	Internal	Not panoramic
[39]	Caries Segmentation	Single-task	DNS (1,500)	OPG	U-Net3+	Dice=0.60	No	Internal	Small dataset
[46]	Caries Segmentation	Single-task	Pediatric OPG (6,075)	OPG	U-Net	F1=0.867	No	Internal	No severity staging
[43]	Multi-disease Classification	Multi-task	Roboflow (1,075)	OPG	CNN Ensemble	Acc=92.9%	No	Internal	No explainability
[15]	Caries Detection	Single-task	CariesXrays (6,000)	OPG	Faster R-CNN	AP50=84.1%	Partial	Internal	Single-disease
[41]	Caries Detection	Single-task	HUNT4 (13k)	Bitewing	YOLOv5	mAP=0.647	No	External	Not panoramic
[3]	Multi-task OPG Analysis	Multi-task	DENTEX (∼1,000)	OPG	YOLOv9-seg	mAP=0.978	No	Internal	Limited pathology
[37]	Tooth + Diagnosis	Multi-task	OdontoAI (1,661) + DENTEX (682)	OPG	Mask DINO + Swin	FROC=0.988, AUC=0.996	No	Internal	Slow pipeline (40s/image)
[42]	Tooth + Caries	Multi-task	935 PAN / 7,304 teeth	OPG	InceptionResNet	Acc=0.94, F1=0.94	No	Internal	No localization
[47]	Caries Segmentation	Single-task	597 PAN	OPG	Faster R-CNN + U-Net	Dice=0.743	No	Internal	Single-disease
[16]	Caries Detection	Single-task	DENTEX + external	OPG	YOLOv8	mAP=0.95	No	External	No segmentation
[21]	Caries Severity	Single-task	1,000 PAN	OPG	Faster R-CNN	Acc=94%	No	Internal	Weak mild detection
[17]	Caries Detection	Multi-stage	500 PAN	OPG	YOLOv5 + U-Net	Recall=0.96	No	Internal	Single-disease
[19]	Bone Loss Classification	Single-task	1,874 OPG	OPG	VGG-16	Acc=89.27%	No	Internal	No localization
[33]	Bone Loss Detection	Single-task	Bitewing (550)	Bitewing	Faster R-CNN + DeepLab	Acc≈85%	No	Internal	Limited modality
[26]	Bone Loss Severity	Single-task	DenPAR (1,000)	IOPA	YOLOv8 + KRCNN	ICC=0.80	No	Internal	Not panoramic
[48]	Bone Loss Segmentation	Single-task	CBNUH (100) + Kaggle (∼600)	OPG	SRGAN + U-Net	Dice=0.91	No	Internal	No detection
[7]	Multi-disease Detection	Single-task	Multi-center panoramic	OPG	YOLOv8	F1=0.60→0.47	No	External	Poor generalization
[49]	Tooth Segmentation	Multi-stage	Tufts + O2PR	OPG	Mask R-CNN + YOLO	AP50=97.3% (Tufts), 98.1% (O2PR)	No	Internal	No disease labels
[35]	Tooth Segmentation Review	Review	20 studies	Multi	CNNs	Sens=0.92	No	Review	High heterogeneity
[2]	Bone Loss Review	Review	30 studies	Multi	CNNs	Acc=84%	No	Review	No standardization
[11]	Deep Learning Survey	Review	260 studies	Multi	Various	—	No	Review	No unified system
[10]	Periodontal Classification	Single-task	Kaggle (∼1,000)	X-ray	PDCNET	Acc=98.39%	Yes	Internal	No localization

performs any caries severity differentiation, and none reports a bone-loss quantity (a millimeter distance, a percentage, or an AAP/EFP stage) on panoramic imaging specifically – Wimalasiri et al.'s ICC-based severity result [26] is the closest analogue in this table, but it is reported on intraoral periapical radiographs, not panoramic ones. As Section 7 discusses, this specific gap is narrowed, though not fully closed on panoramic imaging, by Le et al.'s 2026 keypoint-based bone-loss quantification work [24] and Wang and Li's 2026 caries-staging transformer [23], neither of which was available at the time the base corpus for this table was assembled.

10. DATASETS, BENCHMARKS, AND CLINICAL VALIDATION (C5)

Dataset availability and quality are the most frequently cited limitations across all 70 reviewed papers. The key public benchmarks in the field are as follows:

- **DENTEX**: 1,005 panoramic radiographs with quadrant, tooth, and pathology annotations from three institutions [3], [16], [37].
- **CariesXrays**: 6,000 hospital-grade panoramic images with 13,783 caries bounding boxes [15].
- **Tufts Dental Database**: Approximately 1,000 panoramic radiographs with tooth segmentation masks [49], [53], [54], [61], with a January 2026 benchmark reporting

binary tooth-segmentation Dice = 0.9259 as a current reference point [62].

- **HUNT4 Oral Health Study**: 13,887 bitewing X-rays with six-dentist annotations and a 197-image consensus set [41].
- **UFBA-UESC**: 4,000 panoramic images [21].
- **MMOral** (2025): the largest public panoramic resource identified in this survey – 20,563 annotated radiographs with 1.3 million instruction-following instances, released as a vision-language instruction and benchmark dataset rather than a conventional detection/classification set [38].
- **BRAR-anchored multimodal dataset** (2025): 1,104 patients with inter-rater agreement $\kappa = 0.88$ and a published EfficientNet-B0 baseline (accuracy = 63.3%), intended as a shared benchmark for periodontal bone-loss work [63].

Despite these contributions, roughly 60% of the original 47-paper base corpus relied on private, single-center institutional data. Systematic reviews [1], [2], [12], [34], [35] consistently report I^2 heterogeneity values of 88–96%, which is the primary barrier to reliable evidence synthesis. The performance degradation from internal to external validation documented by Mureşanu et al. [7] – F1 dropping from 0.60 to 0.47 – directly measures the cost of single-center training. The August 2026 update does not reverse this pattern: DentFound's 101,000-patient training corpus, Wang and Li's three-hospital caries corpus, and Balel et al.'s 30,954-image reporting corpus [13], [23], [44] are all private institutional data, even as MMOral and the BRAR dataset show that large *public* resources are now starting to appear.

Two 2026 systematic reviews assess how close the field actually is to clinical deployment rather than benchmark performance alone. Wang et al. [64] document that reported caries-detection AUCs of 0.76–0.98 under internal validation fall to 0.53–0.57 under domain shift, with only partial recovery (0.1–34.2%) from domain-adaptation techniques – a direct, quantified illustration of why this survey treats internally validated metrics with caution throughout. Ardila et al. [65] synthesize 15 studies against six translational-readiness domains (registered prospectively as PROSPERO CRD420261402398) and conclude that most published dental AI systems, across the whole field and not only the studies in this survey, have not been assessed against clinical workflow, cost, or patient-outcome criteria at all – reinforcing Gap 5 in Section 14.

Waqas et al. [66] addressed data scarcity through GAN-based synthetic OPG generation (DCGAN, WGAN) on 4,777 real images. Expert assessors rated synthetic radiographs as realistic above 50% of the time relative to real images, though downstream gains in disease detection accuracy were not demonstrated. Ameli et al. [67] applied YOLOv8 to 1,000 OPGs for automated quality assessment (contrast/density accuracy = 97.9%, artifact accuracy = 87.2%, overall quality = 79.3%), establishing automated quality filtering as a viable preprocessing gate for diagnostic pipelines. Delamare et al. [8] documented heterogeneous quality criteria across 41 panoramic ML studies and argued that standardized quality protocols are a prerequisite for reproducible AI development.

11. Tooth Segmentation and Numbering (C6)

Seven papers address tooth detection, segmentation, and numbering – tasks that are prerequisites for tooth-specific disease attribution. FDI-standard tooth numbering is needed to report that caries is in tooth 26 rather than somewhere in the panoramic image.

Bonfanti-Gris et al. [35] synthesized 20 tooth detection and segmentation studies in a PRISMA/QUADAS-2 meta-analysis, reporting pooled sensitivity = 0.92 and specificity = 0.94 with DOR = 186 – strong aggregate evidence, though $I^2 \approx 96\%$ confirms severe heterogeneity. Zhang et al. [49] proposed a multi-stage system (MobileNetV3 + SADP-Net Mask R-CNN + Swin Transformer + YOLOv5) across three public datasets, achieving AP50 = 97.3% on Tufts and 98.1% on O2PR for tooth instance segmentation. Yaswanth et al. [61] applied Attention-integrated U-Net++ to the Tufts dataset, achieving mAP = 96.4% and Dice = 90%. Adnan and Umer [68] contributed the Orthopantomogram Teeth Segmentation and Numbering Dataset – 250 expert-annotated OPGs on Zenodo – while explicitly noting that disease annotations are absent and that expansion to caries and periodontal labeling is needed.

11.1 2026 Update: Benchmarking, External Validation, and a Direct DENTEX Competitor

Three further C6 studies from the August 2026 update are especially relevant to tooth detection and numbering on panoramic radiographs. Yalım et al. [62] benchmarked ten U-Net encoder backbones on the Tufts Dental Database with 5-fold cross-validation and Friedman/Nemenyi significance testing, reporting Dice ranging from 0.9168 to 0.9259 (EfficientNet-B7 best, EfficientNet-B0 the most practical accuracy-per-parameter operating point) – currently the best-documented published Tufts tooth-segmentation reference point in this survey. Hao et al. [69] trained a YOLOv11 oriented-bounding-box detector for posterior teeth in mixed dentition on a dual-centre corpus of 1,148 radiographs and, critically, evaluated it on a genuine independent external test set of 108 images from a third, unseen centre: mAP@50–95 = 0.904 internally and 0.896 externally, a 0.8-point drop that stands in sharp contrast to the 22-point internal-to-external drop reported elsewhere in this survey [7] and is, to our knowledge, the strongest cross-centre generalization result in the entire corpus. Asif and Khan [70] applied five YOLOv26-segmentation variants to the DENTEX dataset for tooth enumeration and disease segmentation, reporting tooth-enumeration box mAP50 = 0.976 and disease-segmentation box mAP50 = 0.591 – to our knowledge the strongest published DENTEX enumeration result identified in this survey, and the benchmark any future DENTEX-based detection system, including capstone or thesis projects built on this survey's proposed taxonomy, should be reported against.

12. Cross-Study Trends and Observations

This section presents a synthesized analysis of patterns that recur across the 70 reviewed studies. Drawing on the

global comparison in Table IV, several trends emerge across model design, dataset characteristics, task complexity, and deployment challenges.

12.1 Model Architecture Trends

1) Transformer-Based vs. CNN-Based Architectures: Transformer-based architectures such as ViT and Swin Transformer capture global contextual information well, which benefits classification and multi-label diagnosis [37]. In practice, however, CNN-based architectures dominate the reviewed literature, particularly in detection and pipeline-based systems [18]. Their efficiency, lower computational requirements, and reliable behavior on moderately sized datasets explain this dominance.

Transformer-based models require large annotated datasets and exhibit higher inference latency, which reduces their practicality in most clinical environments [37]. CNN-based approaches therefore remain more suitable for deployment under current dataset constraints, despite their representational limitations. A controlled 2025 benchmark directly supports this: across twelve CNN, transformer, and Mamba (state-space) architectures evaluated under matched conditions on caries segmentation, a plain CNN outperformed every transformer and Mamba variant tested [51], suggesting the current data regime in dental radiograph analysis still favors CNN inductive biases. A large private-data counterexample exists – Wang and Li's hierarchical transformer reaches mAP = 87.3% on 3,856 radiographs with explicit caries staging [23] – but it relies on a training corpus roughly two orders of magnitude larger than most CNN studies in this survey, which is consistent with, rather than a refutation of, the data-hungriness argument above.

2) Dominance of YOLO in Detection Tasks: YOLO variants are the most widely used models for caries localization, tooth detection, and bone loss identification across the reviewed literature [3], as reflected in Table IV. Single-pass inference with near real-time performance makes them well suited to clinical applications, where latency matters.

3) Ensemble Methods: Ensemble methods can improve classification robustness by aggregating predictions from multiple models, but the associated increase in computational cost limits their practicality in real-time clinical settings [43].

12.2 Data and Generalization

Dataset size improves model performance but does not guarantee generalization. Large datasets can still suffer from class imbalance, depressing task-specific metrics [18]. More consequentially, models trained on single-center data consistently underperform when tested on external data.

This generalization gap is a structural issue rather than an idiosyncrasy of individual studies. Models trained on panoramic data from a single institution degrade under multi-center conditions [7], and systematic reviews confirm high heterogeneity attributable to differences in imaging protocols, annotation standards, and model architectures [2]. Dataset diversity, multi-center representation, and standardized evaluation protocols are necessary conditions for clinically reliable systems.

12.3 Task Complexity and Learning Paradigm

1) Structural Tasks vs. Disease Detection: Structural tasks such as tooth segmentation and enumeration are relatively low-variance problems; performance is consistently high across architectures [35]. Disease detection – particularly early-stage caries – is substantially harder due to subtle visual features with high intra-class variability.

The implication is that anatomical recognition and pathological detection place fundamentally different demands on model design and data quality. Within the same framework, accuracy typically falls when transitioning from structural to disease detection tasks [3].

2) Multi-Task Learning vs. Single-Task Models: Most reviewed studies adopt single-task formulations; only a small subset attempts multi-task learning, as shown in Table IV. This is a practical limitation, since real-world dental diagnosis requires simultaneous evaluation of multiple conditions. Studies that do combine objectives within a single framework [28] remain rare and are often restricted in scope or modality.

12.4 Deployment and Evaluation Challenges

1) Explainable AI: Explainability appeared in approximately 6.4% of the base-corpus studies, rising to 12.9% with the August 2026 update [13], [28], [29], with the large majority of systems still operating as black-box models. Without interpretable outputs, clinical trust is reduced and regulatory approval becomes difficult. Explainability is therefore a deployment requirement, not an optional feature.

2) Lack of Standardized Evaluation Metrics: Studies report heterogeneous measures – mAP, Dice, IoU, accuracy, F1 – with inconsistent thresholds and experimental setups, making cross-study comparison unreliable. As Table IV shows, this inconsistency complicates benchmarking across models and tasks. Systematic reviews confirm that metric variability is a primary driver of inter-study heterogeneity [35]. Standardized evaluation protocols are needed before the field can make reliable comparative progress.

12.5 Final Observation

CNN-based and YOLO-driven approaches dominate current dental AI, while transformer-based models are emerging but remain resource-intensive. Structural tasks are solved reliably; disease detection – especially early-stage and quantitative assessment – is not. The gap between reported performance on individual benchmarks and the requirements of a clinically deployable pipeline is large, and closing it requires integrated, explainable, generalizable systems rather than continued refinement of isolated sub-tasks.

13. Critical Analysis of Existing Methods

13.1 Metrics and Evaluation Limitations

The reviewed literature lacks standardized evaluation practices. Studies report mAP, Dice coefficient, accuracy, AUC, and F1-score with varying thresholds and experimental setups, making direct comparison across methods unreliable and

obscuring which approaches are genuinely superior [2], [34], [35].

Many studies report strong results under controlled conditions but do not use benchmarking protocols that would permit cross-study comparison. Only a small subset of works adopt consistent evaluation settings. Without unified benchmarks and evaluation standards, the reproducibility and reliability of published results remain fundamentally constrained.

13.2 Dataset Bias and Generalization Issues

A large proportion of reviewed studies use data from a single institution or imaging setup. Single-center datasets fail to capture variability across populations, imaging devices, and acquisition conditions [1], [12], and models trained on them consistently degrade when evaluated externally, as multiple studies confirm [7].

Public datasets such as DENTEX and CariesXrays provide useful baselines but do not cover multi-disease, multi-center clinical variability [3], [15]. Without diverse, standardized, multi-institutional datasets and consistent validation protocols, the generalizability of current systems remains limited.

13.3 Clinical Integration Gap

Most systems are evaluated offline and produce intermediate outputs – bounding boxes, segmentation masks, classification labels – without connecting to real-time clinical decision support. Clinician involvement in model development and evaluation is limited; only a small number of studies assess real-world diagnostic impact in terms of accuracy, treatment planning, or workflow efficiency [32], [33], though the August 2026 update's expert reader studies [13], [44] are a first step in this direction for report-generating systems specifically.

The absence of explainability compounds this problem. Black-box systems without interpretable output create a practical disconnect between technical performance and clinical applicability. Most published systems are not aligned with the requirements of routine clinical use.

13.4 Fragmentation of Existing Approaches

As Table I shows, the majority of reviewed studies target a single task – detection, segmentation, or classification – with only a small fraction exploring integrated multi-task frameworks. Real-world dental diagnosis requires simultaneous evaluation of caries, periodontal status, severity staging, and quantitative measurement. Existing systems do not jointly address these components and therefore cannot reflect the full complexity of a clinical workflow.

No reviewed system operating on panoramic radiographs simultaneously integrates detection, localization, severity staging, quantification, and explainability within a unified framework – DentFound [13] comes closest, integrating four of the five, but omits quantitative severity staging and bone-loss quantification and is not independently reproducible. This is substantially a structural limitation of the field, not merely a property of individual models.

Progress will require a shift from task-specific optimization toward system-level design, where diagnostic components are jointly modeled and aligned with clinical requirements.

These limitations collectively highlight the need for unified, clinically aligned, and generalizable dental AI systems, which are discussed in the following section.

14. Cross-Cutting Gap Analysis

Synthesizing findings across all 70 reviewed papers, five interconnected gaps emerge that consistently constrain clinical translation, though the August 2026 update narrows several of them.

14.1 Gap 1: Single-Disease Focus

Single-disease training is the dominant pattern across all categories. Within the original base corpus, only Chen et al. [28] simultaneously detected both caries and periodontal disease – and exclusively on periapical, not panoramic, radiographs. The August 2026 update identified one panoramic system, DentFound [13], that reports both caries and periodontal findings from a single panoramic exposure. No reviewed study – DentFound included – provides simultaneous caries detection, severity staging, and periodontal bone loss *quantification* within a single, publicly reproducible pipeline. Patients routinely present with co-occurring conditions that must be assessed from a single exposure; no current openly reproducible system fully addresses this.

14.2 Gap 2: Absence of Automated Severity Staging

Of 21 C1 papers, Asadi [21], Bizjak and Robic [16], and Wang and Li [23] distinguish severity beyond binary caries presence. In C2, Wimalasiri et al. [26], Le et al. [24], and Li et al. [25] address bone loss or periodontal severity patterns. The 2018 AAP/EFP classification [22] defines four periodontitis stages; none of the reviewed systems is confirmed to map its staging output explicitly onto that taxonomy on panoramic imaging. Severity staging determines treatment planning, referral, and prognosis. Detection without staging yields limited clinical decision support, and this gap is narrowing rather than closed.

14.3 Gap 3: Explainability at 12.9%

9 of 70 papers now implement some form of XAI output, up from 3 of 47 (6.4%) before the August 2026 update. Regulatory frameworks including the EU AI Act classify AI diagnostic tools as high-risk systems requiring transparency and human oversight – requirements that 61 of the 70 reviewed systems still cannot satisfy. For junior clinicians, explainability is not optional; it is the mechanism through which AI-assisted diagnosis supports learning rather than replacing judgment. The consensus that explainability is necessary for clinical adoption [3], [9], [43], [45] is still not reflected in implementation rates, even after this year-on-year improvement.

14.4 Gap 4: Dataset Fragmentation

Approximately 60% of the original base-corpus studies use private, single-center datasets, and the largest 2026 systems – DentFound, Wang and Li's caries transformer, and Balel et al.'s reporting pipeline [13], [23], [44] – remain private as well, even as public resources such as MMOral [38] begin to emerge. Systematic reviews [1], [2], [12], [34], [35] consistently report $I^2 = 88$–96% heterogeneity as the primary barrier to evidence synthesis. The real cost is quantified by Mureşanu et al. [7]: F1 drops 22% from internal to external validation. The field needs a single large-scale, multi-disease, multi-center, publicly available panoramic OPG dataset with standardized caries and periodontal annotations. MMOral [38] is the first resource in this survey to approach that scale (20,563 panoramic radiographs, publicly released), but it is structured as vision-language instruction and QA data rather than the standardized per-tooth caries and periodontal ground truth this gap calls for, so it narrows rather than closes Gap 4.

14.5 Gap 5: Clinical Workflow Integration

Within the original base corpus, only Schwendicke et al. [32] and Lee et al. [33] evaluated AI against any measure of clinical workflow, cost, or clinician acceptance. This gap is the one most substantially narrowed by the August 2026 update, though not closed at field scale: DentFound [13] was assessed by 12 dentists and radiologists across 595 panoramic radiographs, who rated its generated reports as superior to or comparable with human-written reports; Balel et al. [44] evaluated report quality against an expert gold standard and quantified LLM hallucination rates directly (30 hallucinations across 50 reports for their best local pipeline, versus hallucinations in 100% of reports from commercial cloud LLMs); and Ardila et al. [65] systematically reviewed 15 studies against translational-readiness criteria and found that, field-wide, clinical workflow and outcome evaluation is still the exception rather than the norm. Most systems in this survey still produce bounding boxes or segmentation masks without clinical report generation, severity summaries, or treatment recommendations. The cost analysis by Schwendicke et al. [32] (ICER = -13.9 €/year for AI-assisted caries detection) remains the only economic evaluation in the corpus, even after the 2026 update.

15. Research Roadmap Toward Unified and Clinically Deployable Dental AI Systems

Building on the limitations identified in Section 13, this section outlines a structured roadmap toward unified, clinically deployable dental AI systems.

15.1 Short-Term Goals (1–2 Years)

Public Multi-Disease Panoramic Dataset: The most immediate bottleneck is the absence of a large-scale, standardized panoramic dataset with multi-disease annotations. A consortium-driven effort is needed to produce at least 5,000 panoramic radiographs with consistent annotations for caries (per-tooth, with severity), periodontal bone loss (CEJ–bone crest distance), and tooth numbering (FDI standard).

Datasets such as CariesXrays [15] and DENTEX [3] offer partial solutions but lack comprehensive multi-disease annotations. Extending and integrating them is a practical near-term step.

Explainability as a Standard Component: Explainability should be incorporated by default, using Grad-CAM or attention visualization, with validation against expert annotations to confirm anatomical relevance [10], [18].

15.2 Mid-Term Goals (3–5 Years)

Unified Multi-Disease System for Panoramic Radiographs: The most pressing research direction is a unified multi-disease system that simultaneously detects caries, performs severity staging, quantifies periodontal bone loss, and supports multi-condition diagnosis from a single panoramic radiograph.

This requires multi-head architectures with shared feature representations and task-specific outputs. Individual components exist in the literature, but their integration into a coherent, openly reproducible framework remains unsolved [3], [4], [7], [28]. DentFound [13] demonstrates that the integration itself is achievable at scale; the remaining research task is to reproduce that level of integration with quantitative severity and bone-loss outputs, and with openly released weights and training data rather than a proprietary multi-hospital corpus.

Clinical Alignment with Standardized Diagnostic Frameworks: Future systems should produce outputs that align with established classification standards – ICCMS for caries and the 2018 AAP/EFP classification for periodontal disease [22]. This would bridge the gap between algorithmic output and clinically actionable decisions.

15.3 Long-Term Goals (Clinical Deployment)

Prospective Clinical Validation and Deployment: The field must move beyond retrospective evaluation. Randomized studies assessing diagnostic accuracy, clinician efficiency, and patient outcomes with and without AI assistance are needed, following standardized reporting protocols to allow comparisons across settings.

The cost-effectiveness framework of Schwendicke et al. [32] provides a foundation that should be extended to multi-disease systems and panoramic radiographs. Prospective validation of this kind is essential for both regulatory approval and clinical acceptance.

Final Perspective: *This roadmap describes a path from data standardization and model explainability toward fully integrated, prospectively validated systems – a concrete progression for the next generation of dental AI research.*

16. Conclusion

This survey reviewed 70 deep learning studies (2021–2026) on dental caries detection, periodontal bone loss assessment, and related tasks from panoramic and bitewing dental radiographs, including 23 papers identified in an August

2026 update. Deep learning has reached clinically competitive performance on individual sub-tasks: caries detection achieves 96% recall with Attention U-Net [17]; periodontal landmark detection reaches mAP50 = 0.995 with YOLOv8 [4]; dual-disease classification attains AUC > 98% on periapical X-rays with EfficientNet-B0 and Grad-CAM [28]; AI-assisted caries detection is cost-dominant relative to unaided clinical assessment, with ICER = −13.9 €/year [32]; and, most recently, a vision-language system trained on over 101,000 patients now generates free-text diagnostic reports that clinicians rate as comparable to or better than human-written reports [13].

Five structural gaps persist across the reviewed corpus, each narrowed but not closed by the 2026 update: single-disease models that largely ignore clinical co-occurrence; severity staging and bone-loss quantification that remain rare and are not yet mapped onto standard clinical staging systems; an explainability rate that has risen from 6.4% to 12.9% but still leaves the large majority of systems non-transparent; dataset fragmentation from private single-center collections, only partly offset by the emergence of large public resources such as MMOral; and clinical workflow integration that, outside a small number of 2026 reader studies, remains minimal.

No publicly reproducible system combines caries detection, severity staging, periodontal bone loss quantification, and explainability on panoramic radiographs within a single pipeline. This is a narrower claim than it would have been eighteen months ago: DentFound [13] now demonstrates that detection, numbering, qualitative periodontal findings, post-hoc explainability, and report generation can be integrated into one deployed, clinically evaluated system. What remains unmet – and what this survey's roadmap (Section 15) is scoped to – is a system that additionally reports quantitative, staged severity outputs, and that does so with openly released weights and training data so that its performance claims can be independently audited rather than taken on trust. The technical components exist – YOLO detection, U-Net segmentation, Grad-CAM explainability, EfficientNet classification, transformer-based attention, and now vision-language reporting have each been demonstrated on dental radiographic data. The remaining challenge is measurement-grade integration, open validation, and standardization. Systems that are accurate but not interpretable, not quantitatively staged, or not independently reproducible cannot fully serve the junior clinicians and dental students who stand to benefit most from AI-assisted diagnostic support.

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